

Functional Annotation of PP_4637 (Q88E31) in *Pseudomonas putida* KT2440

1. Summary / Answer

PP_4637 (Q88E31) is not a full methionine synthase, but the N-terminal ($\beta\alpha$)₈-barrel subunit of a *split* cobalamin-independent methionine synthase (MetE, EC 2.1.1.14). In *P. putida* KT2440 the two domains that make up a canonical single-chain MetE enzyme are encoded as two separate genes: **PP_4637** provides the N-terminal folate-polyglutamate-binding / structural barrel (Pfam PF08267, Meth_synt_1), and **PP_2698** provides the C-terminal catalytic barrel (Pfam PF01717, Meth_synt_2) that holds the active-site zinc and homocysteine. Together they catalyze the terminal step of *de novo* methionine biosynthesis — transfer of a methyl group from 5-methyl-tetrahydropteroyl-poly- γ -glutamate to L-homocysteine, yielding L-methionine — in the cytoplasm. PP_4637 itself is **non-catalytic**; its role is to bind the polyglutamylated methyl-folate donor and stabilize the catalytic loop of the PP_2698 subunit.

2. Identity verification (this is the correct target)

- **UniProt:** Q88E31_PSEPK; **locus:** PP_4637; **organism:** *Pseudomonas putida* KT2440 (taxid 160488). ✓ matches the requested target.
- **Length:** 274 aa — far shorter than a canonical MetE (~750 aa).
- **Domain content:** only PF08267 (Meth_synt_1) / InterPro IPR013215 (Cbl-indep_Met_Synth_N), IPR038071 (UROD/MetE-like_sf); SUPFAM SSF51726; PANTHER PTHR30519. UniProt annotates a single "Cobalamin-independent methionine synthase MetE N-terminal" domain (residues 50–232).
- **Missing:** PF01717 (Meth_synt_2), the C-terminal catalytic barrel — confirming this protein is *only* the N-terminal half.
- **Orthology:** KEGG K00549 (EC 2.1.1.14), COG0620, pathway ppu00270 (Cysteine and methionine metabolism).

The gene symbol/description are therefore consistent with the sequence, but the standard "5-methyltetrahydropteroyltriglutamate–homocysteine S-methyltransferase" name **over-calls** the protein: the electronic (IEA) GO/EC annotations describe the full enzyme, whereas Q88E31 is a domain fragment.

3. Key findings and evidence

3.1 PP_4637 is a truncated MetE N-terminal barrel, lacking catalytic machinery

Canonical MetE is a monomer of two face-to-face ($\beta\alpha$)₈ barrels; the **C-terminal barrel** carries the catalytic zinc (ligands His/Cys/Cys — e.g. *A. thaliana* His647/Cys649/Cys733; *E. coli* His641/Cys643/Cys726) that binds and activates homocysteine (Pejchal & Ludwig 2005,

P 15630480 Ferrer et al. 2004,

P 15326182 The N-terminal barrel arose by gene duplication of the C-terminal barrel and its ancestral binding sites are "obliterated" (

P 15630480 Q88E31 contains only this N-terminal barrel (PF08267) and none of the zinc/homocysteine catalytic residues — so it cannot, on its own, methylate homocysteine.

3.2 *P. putida* KT2440 has a *split* MetE (PP_4637 + PP_2698)

A genome-wide survey found **no single-chain ~750-aa MetE**. Instead: - **PP_4637 / Q88E31** (274 aa, PF08267, N-terminal barrel) — KEGG K00549 / EC 2.1.1.14. - **PP_2698 / Q88JF1** (344 aa, PF01717 "MetE C-terminal/archaeal" barrel; conserved catalytic Cys/His including a C-terminal CxxC "CTNCG" zinc motif) — also KEGG K00549 / EC 2.1.1.14.

Both are annotated as the same MetE reaction. The two loci are distant (PP_4637 $\approx$ 5.26 Mb; PP_2698 $\approx$ 3.09 Mb; not an operon), yet in the **STRING functional-association network the PP_4637–metE(PP_2698) edge is the single highest-scoring interaction of the whole methionine module (0.989)** — strong evidence that the two barrels partner to reconstitute the enzyme. PP_4637 is also tightly coupled to the rest of the sulfur/methionine module (metH, metK/SAM synthase, ahcY/SAH hydrolase, metB, metZ, hom, metG).

A separate, cobalamin-DEPENDENT methionine synthase, **MetH (PP_2375 / Q88KB5, 1235 aa, K00548 / EC 2.1.1.13)**, provides the B12-dependent route — so *P. putida* has both the B12-independent (split MetE) and B12-dependent (MetH) branches.

3.3 Genomic context, one-to-one pairing, and fold quality (corroborating evidence)

- **No fused MetE in KT2440:** the strain has exactly **one** PF08267 protein (PP_4637), exactly **one** PF01717 protein (PP_2698), and **zero** fused (both-domain) MetE proteins — so these two genes constitute the organism's entire cobalamin-independent methionine-synthase capacity. (Most other *Pseudomonas* carry a fused single-chain MetE, making KT2440's split arrangement notable.)
- **metF–metE co-localization:** PP_4637's immediate chromosomal neighbor **PP_4638 is methylenetetrahydrofolate reductase (MetF/MTHFR)** — the enzyme that *produces* 5-methyl-tetrahydrofolate, the exact methyl-donor substrate of MetE. The next gene, PP_4639, is a P-loop GTPase-dependent **zinc chaperone** (relevant to the zinc-dependent MetE reaction). This classic *metF/metE* juxtaposition is strong functional-context evidence.
- **Confidently folded barrel:** the AlphaFold model AF-Q88E31-F1 has **mean pLDDT 91.8** (95% of residues >70, 77% >90) across all 274 residues — a well-ordered ($\beta\alpha$)₈ barrel, i.e. a genuine functional domain, not a decaying pseudogene fragment.
- **STRING identity confirmed:** the STRING "metE" node that partners PP_4637 (edge 0.989) resolves to locus **160488.PP_2698**, and "metH" to PP_2375.

3.4 Direct sequence proof of the split (each subunit = one half of a canonical MetE)

Smith–Waterman alignment (BLOSUM62) against *E. coli* MetE (P25665, 753 aa) maps the two *P. putida* proteins to **non-overlapping halves** of one canonical MetE: - **PP_4637 → E. coli MetE residues 134–335** (its own residues 51–255), score 454, **47% identity over 202 residues** — the **N-terminal barrel**. - **PP_2698 → E. coli MetE residues 427–631** (its own residues 6–210), score 260, **33% identity over 204 residues** — the **C-terminal catalytic barrel**.

Both retain the barrel-diagnostic DExP motif (PP_4637 'VQIDEP'; PP_2698 'IQFDEP'). This is direct, quantitative evidence — beyond family-level annotation — that KT2440's MetE is a genuine split of the canonical two-barrel enzyme.

3.5 Precise role, localization, and pathway

- **Reaction (reconstituted enzyme):** 5-methyl-H₄-pteroyl-poly-γ-glutamate + L-homocysteine → L-methionine + H₄-pteroyl-poly-γ-glutamate (EC 2.1.1.14); no intermediate methyl carrier, cobalamin-independent.
- **PP_4637's specific contribution:** the N-terminal barrel **binds the poly-γ-glutamate tail of the methyl-folate substrate and stabilizes the catalytic (folate-binding) loop of the C-terminal domain** ("a structural role of the N-terminal domain ... in the stabilization of loop 507–529 and in the interaction with the poly-glutamate chain of CH₃-H₄PteGlu_n" —

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Catalysis (methyl transfer, zinc-activated thiolate) occurs on the **C-terminal (PP_2698) subunit**.

- **Substrate specificity note:** MetE strongly prefers **polyglutamylated** methyl-tetrahydrofolate; the N-terminal barrel is a key determinant of that polyglutamate preference.
- **Localization:** cytoplasmic — Q88E31 is a soluble 274-aa protein with no signal peptide or transmembrane segment, consistent with the cytosolic MetE family.
- **Pathway:** final step of **de novo L-methionine biosynthesis** within sulfur assimilation; methionine feeds **S-adenosylmethionine (SAM)** synthesis (metK) and one-carbon/folate cycling (regenerated tetrahydrofolate; SAH recycled via ahcY).

4. Hypotheses — supported and refuted

Supported - PP_4637 is the N-terminal barrel of MetE (domain, orthology, structure). ✓ - *P. putida* KT2440 uses a two-gene (split) MetE; PP_4637 partners with the PP_2698 catalytic subunit (STRING top edge 0.989; shared K00549). ✓ - PP_4637 acts in the cytoplasm in methionine/SAM biosynthesis. ✓ (family localization + module context)

Refuted / corrected - PP_4637 is a standalone, catalytically complete 5-methyl-THF-triglutamate-homocysteine methyltransferase. ✗ Refuted — it lacks the entire C-terminal catalytic barrel and the zinc/homocysteine site; the database EC/GO calls reflect family membership, not verified activity of this fragment.

5. Limitations and future directions

- Evidence is bioinformatic/structural-homology based; there is **no direct biochemical or genetic study of PP_4637 itself**. No PubMed literature exists for this specific *P. putida* protein.
- Direct physical association of PP_4637 with PP_2698 is inferred from STRING co-occurrence/ context, not demonstrated experimentally. Whether the two barrels form a stable heterocomplex, or whether PP_2698 functions as an autonomous "archaeal-type" C-terminal-only synthase with PP_4637 as an auxiliary folate-binding factor, remains to be tested.
- **Suggested experiments:** (i) Δ PP_4637 and Δ PP_2698 growth phenotypes on minimal medium $\pm$ methionine $\pm$ vitamin B12 (to test whether the split MetE is the sole B12-independent route); (ii) co-purification / pull-down of PP_4637 with PP_2698; (iii) in vitro reconstitution of EC 2.1.1.14 activity from the two subunits with mono- vs poly-glutamate methyl-folate; (iv) AlphaFold-Multimer modeling of the PP_4637:PP_2698 heterodimer to confirm the face-to-face double-barrel interface.

6. Primary references

- Pejchal R, Ludwig ML (2005) *Cobalamin-independent methionine synthase (MetE): a face-to-face double barrel that evolved by gene duplication*. PLoS Biol. PMID **15630480**.
- Ferrer J-L, Ravanel S, Robert M, Dumas R (2004) *Crystal structures of cobalamin-independent methionine synthase complexed with zinc, homocysteine, and methyltetrahydrofolate*. J Biol Chem. PMID **15326182**.
- Fu T-M et al. (2011) *Crystal structures of cobalamin-independent methionine synthase (MetE) from Streptococcus mutans: a dynamic zinc-inversion model*. PMID **21840320**.
- Database sources: UniProt Q88E31 / Q88JF1 / Q88KB5; KEGG ppu:PP_4637, PP_2698, PP_2375; Pfam PF08267/PF01717; InterPro IPR013215/IPR038071; STRING (taxid 160488).